

# LIST 3: EXPANDING FUNCTIONS INTO THE TRIGONOMETRIC FOURIER SERIES

Electronics and Telecommunications; May 27, 2026 Kinga McInerney

**Exercise 1** Check if those functions satisfy Dirichlet Conditions. If yes, expand them into the trigonometric Fourier series and discuss type of convergence. Notice that all of these functions are  $2\pi$  periodic.

1.  $f(x) = x^2, x \in \langle -\pi; \pi \rangle$

3.  $f(x) = \sin ax, x \in (-\pi; \pi), a \in \mathbb{R} \setminus \mathbb{Z}$

2.  $f(x) = x^3, x \in \langle -\pi; \pi \rangle$

4.  $f(x) = \cos ax, x \in (-\pi; \pi), a \in \mathbb{R} \setminus \mathbb{Z}$

**Exercise 2** Calculate trigonometric Fourier coefficients for the following  $2\pi$  periodic functions.

1.  $f(x) = \begin{cases} -x, & x \in \langle -\pi; 0 \rangle \\ 0, & x = -\pi \text{ and } x = \pi \\ x, & x \in (0; \pi) \end{cases}$

5.  $f(x) = \begin{cases} -c, & x \in (-\pi; 0), (c > 0) \\ 0, & x = 0 \\ c, & x \in (0; \pi) \end{cases}$

2.  $f(x) = \begin{cases} 0, & x \in (-\pi; 0) \\ x, & x \in (0; \pi) \end{cases}$

6.  $f(x) = \begin{cases} 0, & x \in (-\pi; 0) \\ \sin x, & x \in (0; \pi) \end{cases}$

3.  $f(x) = \begin{cases} 2x, & x \in (-\pi; 0) \\ 6x, & x \in (0; \pi) \end{cases}$

7.  $f(x) = \begin{cases} x, & x \in (-\pi; \pi), \\ 0, & x = -\pi \text{ and } x = \pi \end{cases}$

4.  $f(x) = \begin{cases} -\cos x, & x \in (-\pi; 0) \\ 0, & x = 0 \\ \cos x, & x \in (0; \pi) \end{cases}$

8.  $f(x) = \begin{cases} 1, & x \in (0, +\infty) \\ \frac{1}{2}, & x = 0 \\ 0, & x \in (-\infty; 0) \end{cases}$

**Exercise 3** Expand  $f(x) = \begin{cases} 0, & x \in (0; \frac{\pi}{2}) \\ \pi, & x \in (\frac{\pi}{2}; \pi) \end{cases}$  into the sinus Fourier series.

**Exercise 4** Expand  $f(x) = \begin{cases} 0, & x \in (0; \frac{\pi}{2}) \\ x, & x \in (\frac{\pi}{2}; \pi) \end{cases}$  into the cosine Fourier series.

**Exercise 5** Expand  $f(x) = \begin{cases} x, & x \in (0; \frac{\pi}{2}) \\ 0, & x \in (\frac{\pi}{2}; \pi) \end{cases}$  into the sinus Fourier series.

**Exercise 6** Expand  $f(x) = \begin{cases} x, & x \in (0; \frac{\pi}{2}) \\ 0, & x \in (\frac{\pi}{2}; \pi) \end{cases}$  into the cosine Fourier series.

**Exercise 7** Expand  $f(x) = x, x \in (-\pi; \pi)$  into the Fourier series. Using the expansion calculate the sums of the following number series  $\sum_{n=1}^{+\infty} \frac{1}{n^2}, \sum_{n=1}^{+\infty} \frac{(-1)^n}{n^2}, \sum_{n=1}^{+\infty} \frac{1}{(2n-1)^2}$ .

**Exercise 8** Expand  $f(x) = \frac{\pi}{4}, x \in (-\pi; \pi)$  into the Fourier series. Using the expansion calculate the sums  $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

**Exercise 9** Expand  $f(x) = x^4, x \in (-\pi; \pi)$  into the Fourier series. Using the expansion calculate the sum  $\sum_{n=1}^{+\infty} \frac{1}{n^4}$ .